Contents

1 Smooth Maps

1.1 Smooth Functions and Smooth Maps

1.1.1 Smooth Maps Between Manifolds

Definition 1.1.1 ▸ Smooth Maps Between Manifolds

Let M, N be smooth manifolds, and let $F : M \to N$ be any map. We say that F is a ***smooth map*** if for every $p \in M$, there exist smooth charts (U, φ) containing p and (V, ψ) containing $F(p)$ such that $F(U) \subseteq V$ and the composite map $\psi \circ F \circ \varphi^{-1}$ is smooth from $\varphi(U)$ to $\psi(V)$.

Remark.

- By taking $N = V = \mathbb{R}^k$ and $\psi = \mathrm{Id} : \mathbb{R}^k \to \mathbb{R}^k$, we get the definition of smoothness of real-valued or vector-valed functions.

1.2 Partitions of Unity

Definition 1.2.1 ▸ Support

If f is any real-valued or vector-valued function on a topological space M, the ***support of f***, denoted by $\operatorname{supp} f$, is the closure of the set of points where f is nonzero:

$$\operatorname{supp} f = \overline{\{p \in M : f(p) \neq 0\}}.$$

- If $\operatorname{supp} f$ is contained in some set $U \subseteq M$, we say that f is ***supported in U***.
- A function f is said to be ***compactly supported*** if $\operatorname{supp} f$ is a compact set, denoted by $f \in C_0^\infty(M)$.

Remark.
Clearly, every function on a compact space is compactly supported.

Lemma 1.2.2

The function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \leq 0, \end{cases}$$

is smooth.

Lemma 1.2.3

Given any real numbers r_1 and r_2 such that $r_1 < r_2$, there exists a smooth function $h : \mathbb{R} \to \mathbb{R}$ such that $h(t) \equiv 1$ for $t \leq r_1$, $0 < h(t) < 1$ for $r_1 < t < r_2$, and $h(t) \equiv 0$ for $t \geq r_2$.

Proof. Let f be the function of the previous lemma, and set

$$h(t) = \frac{f(r_2 - t)}{f(r_2 - t) + f(t - r_1)}.$$

Note that the denominator is positive for all t, because at least one of the expressions $r_2 - t$ and $t - r_1$ is always positive. The desired properties of h follow easily from those of f. $\square$

Lemma 1.2.4 ▶ Bump Function on $\mathbb{R}^n$

Given any positive real numbers $r_1 < r_2$, there is a smooth function $H : \mathbb{R}^n \to \mathbb{R}$ such that $H \equiv 1$ on $\bar{B}_{r_1}(0)$, $0 < H(x) < 1$ for all $x \in B_{r_2}(0) \setminus \bar{B}_{r_1}(0)$, and $H \equiv 0$ on $\mathbb{R}^n \setminus B_{r_2}(0)$.

Proof. Just set $H(x) = h(|x|)$, where h is the function of the preceding lemma. Clearly, H is smooth on $\mathbb{R}^n \setminus \{0\}$, because it is a composition of smooth functions there. Since it is identically equal to 1 on $B_{r_1}(0)$, it is smooth there too. $\square$

Theorem 1.2.5 ▶ Existence of Bump Function on Manifold

Let M be a smooth manifold, $K \subseteq M$ a compact set. $U \subseteq M$ is a open set such that $K \subseteq U$. Then there exists $\varphi \in C_0^\infty(M)$, such that $\operatorname{supp}\varphi \subseteq U$, $0 \leq \varphi \leq 1$ and $\varphi \equiv 1$ on K.

Proof. For each $x \in K$, take a chart (φ_x, U_x, V_x) centering x. We may assume that $V_x \subseteq B_3(0)$ and $U_x \subseteq U$ by contracting when necessary . Let $\tilde{U}_x := \varphi_x^{-1}(B_1(0))$. We define

$$f_x(p) := \begin{cases} H \circ \varphi_x(p), & p \in U_x \\ 0, & p \notin U_x \end{cases}$$

Where H is defined in Lemma 1.2.4 and we take $r_1 = 1, r_2 = 2$. It is straightforward to verify that $f_x \equiv 1$ on $\tilde{U}_x$ and $f_x \equiv 0$ on U_x^c, $\operatorname{supp} f_x = \varphi_x^{-1}\left(\overline{B_2(0)}\right) \subseteq U_x$. Since the homeomorphism keeps the closure $\varphi_x^{-1}\left(\overline{B_2(0)}\right) = \overline{\varphi_x^{-1}(B_2(0))}$.

Now K has a open cover $\{\tilde{U}_x\}$. Since K is compact, there are finitely many $\tilde{U}_{x_i}$ such that

$$\bigcup_{i=1}^{N} \tilde{U}_{x_i} \supseteq K$$

We define

$$\psi := \sum_{i=1}^{N} f_{x_i}$$

Then $\psi \geq 1$ on K, $\operatorname{supp} \psi \subseteq U$.

Let $\rho : \mathbb{R} \to [0,1]$ be a smooth function such that

$$\rho(t) = \begin{cases} 0, & t \leq 1/2 \\ 1, & t \geq 1 \end{cases}$$

and ρ is monotonically increasing on $(1/2, 1)$. Now, we define our final function $\varphi : M \to \mathbb{R}$ as the composition

$$\varphi := \rho \circ \psi$$

□

Definition 1.2.6 ▸ Partitions of Unity

Suppose M is a topological space, and let $\mathcal{X} = (X_\alpha)_{\alpha \in A}$ be an arbitrary open cover of M, indexed by a set A. A ***partition of unity subordinate to*** $\mathcal{X}$ is an indexed family $(\psi_\alpha)_{\alpha \in A}$ of continuous functions $\psi_\alpha : M \to \mathbb{R}$ with the following properties:

1. $0 \le \psi_\alpha(x) \le 1$ for all $\alpha \in A$ and all $x \in M$.
2. $\operatorname{supp} \psi_\alpha \subseteq X_\alpha$ for each $\alpha \in A$.
3. The family of supports $(\operatorname{supp} \psi_\alpha)_{\alpha \in A}$ is locally finite, meaning that every point has a neighborhood that intersects $\operatorname{supp} \psi_\alpha$ for only finitely many values of α.
4. $\sum_{\alpha \in A} \psi_\alpha(x) = 1$ for all $x \in M$.

If M is a smooth manifold with or without boundary, a ***smooth partition of unity*** is one for which each of the functions ψ_α is smooth.

Remark.

- Because of the local finiteness condition (3), the sum in (4) actually has only finitely many nonzero terms in a neighborhood of each point, so there is no issue of convergence.
- For a manifold M, M is Lindelöf Space, we can always assmue that $(X_\alpha)_{\alpha \in A}$ is a countable colletction, thus the $(\psi_\alpha)_{\alpha \in A}$ is as well .

Theorem 1.2.7 ▸ Existence of Partitions of Unity

Suppose M is a smooth manifold with or without boundary, and $\mathcal{X} = (X_\alpha)_{\alpha \in A}$ is any indexed open cover of M. Then there exists a smooth partition of unity subordinate to $\mathcal{X}$.

Applications

Proposition 1.2.8 ▸ Bump function Support on a Closed Set

Let M be a manifold, $A \subseteq M$ be a closed set, $U \supseteq A$ is an open set. Then there exists $\varphi \in C_0^\infty(M)$ such that $0 \le \varphi \le 1$, and $\varphi \equiv 1$ on A .

Proof. $\{U, M \setminus A\}$ is a open cover of M. Then there exists a partition of unity $\{\rho_1, \rho_2\}$ subordinate to it. Then ρ_1 is just the function we need. □

Theorem 1.2.9 ▸ Stone-Weierstrass

Let X be a compact Hausdorff Space. $\mathcal{A} \subseteq C(X, \mathbb{R})$ is a subalgebra . Then $\mathcal{A}$ is dense in $C(X, \mathbb{R})$, if and only if it satisfies

1. Nowhere vanishing: For each $x \in X$, there exists $g \in \mathcal{A}$ such that $g(x) \neq 0$.
2. Separates points: For distinct $x, y \in X$, there exists $f \in \mathcal{A}$ such that

$f(x) \neq g(x)$.

Theorem 1.2.10 ▶ Whitney

Let M be a smooth manifold. For each continuous function $g \in: M \to \mathbb{R}$, $\delta : M \to \mathbb{R}_+$, there exists a smooth function $f : M \to \mathbb{R}$, such that

$$|f(p) - g(p)| < \delta(p), \quad \forall p \in M$$

Theorem 1.2.11 ▶ Whitney Extension

Let M be a smooth manifold. $A \subseteq M$ is a closed set. For continuous $g : M \to \mathbb{R}$, $\delta : M \to \mathbb{R}_+$, where g is smooth on A. Then there exists smooth function $f : M \to \mathbb{R}$, such that

1. $$f(p) = g(p), \quad \forall p \in A$$
2. $$|f(p) - g(p)| < \delta(p), \quad \forall p \in M$$